

Customer Churn Prediction Using Machine Learning

Final Report

Introduction

This project aims to develop a machine learning model to predict customer churn using the Telco Customer Churn dataset.

Customer churn prediction helps companies identify customers who are likely to leave their services, allowing businesses to take proactive actions to improve customer retention.

The project follows a complete machine learning workflow, including data preprocessing, feature engineering, model training, evaluation, optimization, and prediction.

Methodology

The project was completed using the following steps:

- Data loading and exploration.
- Data cleaning and preprocessing.
- Feature engineering and categorical feature encoding.
- Splitting the dataset into training and testing sets.
- Training a Random Forest classifier.
- Evaluating the model using Accuracy, Confusion Matrix, and Classification Report.
- Optimizing the model using GridSearchCV.
- Identifying the most important features influencing customer churn.
- Predicting the churn status of a new customer using the optimized model.

Results

The initial Random Forest model achieved an accuracy of **78.50%**.

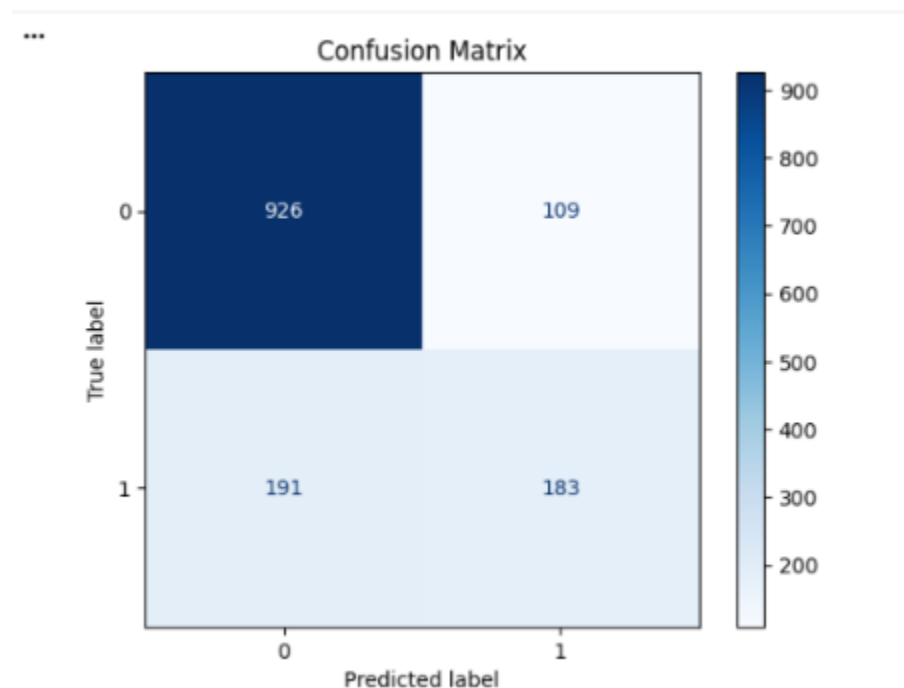
After applying GridSearchCV, the optimized model achieved an accuracy of **80.62%**, showing an improvement in predictive performance.

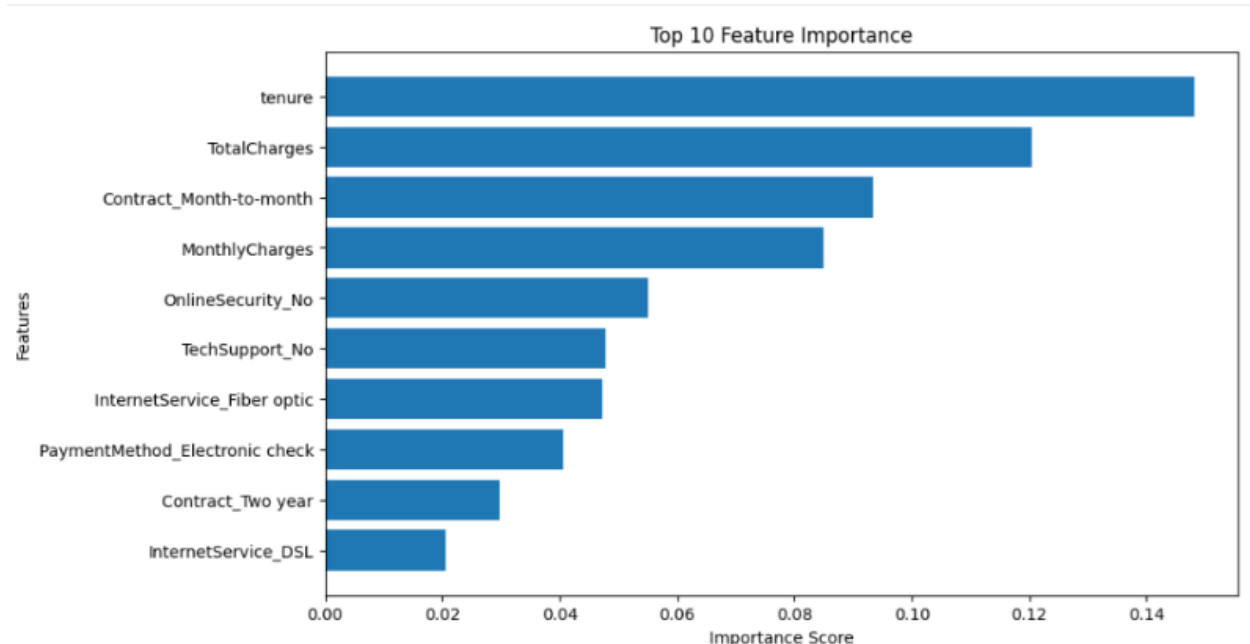
The confusion matrix demonstrated that the model performed well in identifying customers who remained with the company. However, predicting customers who churned remained more challenging because the dataset contains fewer churn cases.

Feature importance analysis showed that the most influential variables were:

- Tenure
- Total Charges
- Contract Type
- Monthly Charges
- Online Security

These variables had the greatest impact on predicting customer churn.





Conclusion

This project successfully developed and optimized a Random Forest model for customer churn prediction.

Hyperparameter tuning improved the model accuracy from **78.50%** to **80.62%**. Feature importance analysis provided valuable insights into the factors influencing customer churn, while the optimized model successfully predicted the churn status of a new customer.

Overall, this project demonstrated the complete machine learning workflow, from data preprocessing to model optimization and prediction, providing practical experience in solving a real-world business problem.